

Math527 – Numerical Analysis

Research Projects

You must choose one from the following projects as your course project.

- Proj 1.** Consider the complex polynomial $z^3 - 1$, whose zeros are the three cube roots of unity. Generate a picture showing three basins of attraction in the complex plane in the square region defined by $-1 \leq \text{Real}(z) \leq 1$ and $-1 \leq \text{Imaginary}(z) \leq 1$. To do this, use a mesh of 1000×1000 pixels inside the square. The center point of each pixel is used to start the iteration of Newton's method (See Section 3.2). Assign a particular basin color to each pixel if convergence to a root is obtained with $n_{max} = 10$ iterations. The criterion for convergence is to check both $|z_{n+1} - z_n| < \epsilon$ and $|z^3 - 1| < \epsilon$ with a small value such as $\epsilon = 10^{-4}$ as well as a maximum number of iterations. Also repeat your work for the polynomial $z^4 - 1 = 0$.
- Proj 2.** Design and carry out computer experiments using the Romberg algorithm based on the trapezoidal formula for numerical integration (See Section 5.3). For a function that possesses many continuous derivatives on the interval, the method should work well. Test your codes on the following integrals and explain the results (errors and convergences):

$$\int_0^1 (1+x)^{-1} dx, \quad \int_{-1}^1 e^x dx, \quad \int_{-10}^{10} (1+x^2)^{-1} dx, \quad \int_0^1 \sqrt{x} dx$$

- Proj 3.** Describe and carry out the programming project for the Subbotin quadratic spline (See Section 9.2). Test your codes by use of the Subbotin quadratic spline in fitting a curve to the following set of data:

x	0.0	0.6	1.5	1.7	1.9	2.1	2.3	2.6	2.8	3.0	3.6	4.7	5.2
y	-0.8	-0.34	0.59	0.59	0.23	0.1	0.28	1.03	1.5	1.44	0.74	-0.82	-1.27

Draw the fitting curve.

- Proj 4.** Carry out some experiments in data smoothing. Start with certain polynomial of degree n (arbitrarily chosen). Compute 100 values of this polynomial at random points in the interval $[-1, 1]$. Perturb these values by adding random numbers chosen from a small interval such as $[-1/8, 1/8]$. Try to recover the polynomial from these polynomial from these perturbed values by using the method of least squares. You may use the Chebyshev polynomials $\{T_0(x), T_1(x), \dots, T_n(x)\}$ or the polynomials $\{1, x, x^2, \dots, x^n\}$ as the basis functions in your implementation. Test your code for $n = 6, 7, 8$.